

Chapter Summary

Key Terms

8.1 Gathering and Organizing Data

- sample
- units
- data
- population
- simple random sample
- systematic random sample
- stratified sample
- cluster sample
- quantitative data
- categorical data
- categorical frequency Distribution
- binned frequency distribution

8.2 Visualizing Data

- proportion
- bar chart
- pie chart
- stem-and-leaf plot
- distribution (of quantitative data)
- histogram
- bar chart for labeled data

8.3 Mean, Median and Mode

- mode
- bimodal
- median
- mean

8.4 Range and Standard Deviation

- range
- standard deviation

8.5 Percentiles

- percentile
- quartile
- quintile
- quantile

8.6 The Normal Distribution

- normal distribution
- 68-95-99.7 Rule
- standardized score (z -score)

8.8 Scatter Plots, Correlation, and Regression Lines

- response variable (dependent variable)
- explanatory variable (independent variable)
- scatter plot
- positive linear relationship
- negative linear relationship
- correlation coefficient
- regression line (least-squares line, line of best fit)

Key Concepts

8.1 Gathering and Organizing Data

- Categorical data places units into groups (categories), while quantitative data is a numerical measure of a property of a unit.
- The sampling method for a study depends on the way that randomization is used to select units for the sample.
- Frequency distributions help to summarize data by counting the number of units that fall into a particular category or range of quantitative values.

8.2 Visualizing Data

- Categorical data can be visualized using pie charts or bar charts; quantitative data can be visualized using stem-and-leaf plots or histograms.
- Areas in pie charts and bar charts represent proportions of the data falling into a particular category, while areas in histograms represent proportions of the data that fall into a given range of data values (or “bins”). Stem-and-leaf plots are visual representations of entire datasets.
- By manipulating the axes, changing widths of bars, or making bad choices for bins, we can create data visualizations that misrepresent the distribution of data.

8.3 Mean, Median and Mode

- The mode of a dataset is the value that appears the most frequently. The median is a value that is greater than or equal to no more than 50% of the data and less than or equal to no more than 50% of the data. The mean is the sum of all the data values, divided by the number of units in the dataset.
- The median of a dataset is not affected by outliers, but the mean will be biased toward outliers. This distinction might affect which measure of centrality is used to summarize a dataset.

8.4 Range and Standard Deviation

- The range of a dataset is the difference between its largest and smallest values. The standard deviation is approximately the mean difference (in absolute value) that individual units fall from the mean of the dataset.

8.5 Percentiles

- The percentile rank of a data value is the percentage of all values in the dataset that are less than or equal to the given value.

8.6 The Normal Distribution

- Normally distributed data follow a bell-shaped, symmetrical distribution.
- The mean of normally distributed data falls at the peak of the distribution. The standard deviation of normally-distributed data is the distance from the peak to either of the inflection points.
- Data that are normally distributed follow the 68-95-99.7 Rule, which says that approximately 68% of the data fall within one standard deviation of the mean, 95% fall within two standard deviations, and 99.7% fall within three standard deviations.
- The z -score for a data value is the number of standard deviations that value falls above (or below, if the z -score is negative) the mean.
- We can use the normal distribution to estimate percentiles.

8.7 Applications of the Normal Distribution

- We can use z -scores to compare data values from different datasets.

8.8 Scatter Plots, Correlation, and Regression Lines

- If one variable affects the value of another variable, we say the first is an explanatory variable and the second is a response variable.
- Scatter plots place a point in the xy -plane for each unit in the dataset. The x -value is the value of the explanatory variable, and the y -value is the value of the response variable.
- The correlation coefficient r gives us information about the strength and direction of the relationship between two variables. If r is positive, the relationship is positive: an increase in the value of the explanatory variable tends to correspond to an **increase** in the value of the response variable. If r is negative, the relationship is negative: an increase in the value of the explanatory variable tends to correspond to a **decrease** in the value of the response variable. Values of r that are close to 0 indicate weak relationships, while values close to -1 or 1 indicate strong

relationships.

- The regression line for a relationship between two variables is the line that best represents the data. It can be used to predict values of the response variable for a given value of the explanatory variable.

Videos

8.2 Visualizing Data

- [Make a Simple Bar Graph in Google Sheets \(https://openstax.org/r/Bar-Graphs\)](https://openstax.org/r/Bar-Graphs)
- [Create Pie Charts Using Google Sheets \(https://openstax.org/r/Pie-Charts\)](https://openstax.org/r/Pie-Charts)
- [Make a Histogram Using Google Sheets \(https://openstax.org/r/Histograms\)](https://openstax.org/r/Histograms)
- [How to Spot a Misleading Graph \(https://openstax.org/r/Misleading-Graphs\)](https://openstax.org/r/Misleading-Graphs)

8.3 Mean, Median and Mode

- [Compute Measures of Centrality Using Google Sheets \(https://openstax.org/r/Computing-Measures-of-Centrality\)](https://openstax.org/r/Computing-Measures-of-Centrality)

8.4 Range and Standard Deviation

- [Find the Minimum and Maximum Using Google Sheets \(https://openstax.org/r/Finding-Minimum-and-Maximum\)](https://openstax.org/r/Finding-Minimum-and-Maximum)
- [Find the Standard Deviation Using Google Sheets \(https://openstax.org/r/Finding-Standard-Deviation\)](https://openstax.org/r/Finding-Standard-Deviation)

8.5 Percentiles

- [Using RANK, PERCENTRANK, and PERCENTILE in Google Sheets \(https://openstax.org/r/Using-RANK-PERCENTRANK-and-PERCENTILE\)](https://openstax.org/r/Using-RANK-PERCENTRANK-and-PERCENTILE)

8.8 Scatter Plots, Correlation, and Regression Lines

- [Making Scatter Plots in Google Sheets \(https://openstax.org/r/Scatter-Plots\)](https://openstax.org/r/Scatter-Plots)

Formula Review

8.3 Mean, Median and Mode

Suppose we have a set of data with n values, ordered from smallest to largest. If n is odd, then the median is the data value at position $\frac{n+1}{2}$. If n is even, then we find the values at positions $\frac{n}{2}$ and $\frac{n}{2} + 1$. If those values are named a and b , then the median is defined to be $\frac{a+b}{2}$.

8.4 Range and Standard Deviation

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

Here, s is the standard deviation, x represents each data value, $\bar{x}$ is the mean of the data values, n is the number of data values, and the capital sigma (Σ) indicates that we take a sum.

8.6 The Normal Distribution

If x is a member of a normally distributed dataset with mean μ and standard deviation σ , then the standardized score for x is

$$z = \frac{x - \mu}{\sigma}.$$

If you know a z -score but not the original data value x , you can find it by solving the previous equation for x :

$$x = \mu + z \times \sigma.$$

8.8 Scatter Plots, Correlation, and Regression Lines

If a line has slope m and passes through a point (x_0, y_0) , then the point-slope form of the equation of the line is:

$$y = m(x - x_0) + y_0$$

Suppose x and y are explanatory and response datasets that have a linear relationship. If their means are $\bar{x}$ and $\bar{y}$ respectively, their standard deviations are s_x and s_y respectively, and their correlation coefficient is r , then the equation of the regression line is:

$$y = r \left(\frac{s_y}{s_x} \right) (x - \bar{x}) + \bar{y}.$$

Projects

1. Browse through some news websites to find five stories that report on data and include data visualizations. Can you tell from the report how the data were collected? Was randomization used? Are the visualizations appropriate for the data? Are the visualizations presented in a way that might bias the reader?
2. We discussed three measures of centrality in this chapter: the mode, the median, and the mean. In a broader context, the mean as we discussed it is more properly called the *arithmetic mean*, to distinguish it from other types of means. Examples of these include the geometric mean, harmonic mean, truncated mean, and weighted mean. How are these computed? How do they compare to the arithmetic mean? In what situations would each of these be preferred to the arithmetic mean?
3. *Simpson's Paradox* is a statistical phenomenon that can sometimes appear when we observe a relationship within several subgroups of a population, but when the data for all the groups are analyzed all together, the opposite relationship appears. Find some examples of Simpson's Paradox in real-world situations, and write a paragraph or two that would explain the concept to someone who had never studied statistics before.

Chapter Review

Gathering and Organizing Data

Decide whether randomization is being used in the selection of these samples. If it is, identify the type of random sample (simple, systematic, cluster, or stratified).

- College students want to gauge interest in a new club they're thinking about starting. They choose three residence halls at random and slide a survey form under the door to every room.
- The managers of a campus dining hall want feedback on a new dish that is being served, so they ask the first 10 people who choose that dish what they think of it.
- The administration of a large university system with five campuses wants to administer a survey to students in the system. The administration chooses 50 students at random from each campus; the 250 students selected form the sample.
- A sample of college students were asked how many of their meals were provided by a campus dining hall on the previous day. Here are the results:

0	1	0	3	2	3	3
2	2	0	2	1	1	0
0	1	2	0	3	0	1
3	2	3	0	1	3	0

Create a frequency distribution for these data.

Visualizing Data

- Create a bar chart to visualize the data in question 4.
- Use the data in “MBB2019” (https://openstax.org/r/Chapter8_Data-Sets) on every school that fielded an NCAA Division I men's basketball team in the 2018–2019 season, to create a histogram of the number of free throws made by each team (FT).

Mean, Median and Mode

- The data below show the top ten scorers (by points per game) in the National Basketball Association for the 2018–2019 season. Find the mode, median, and mean age of these players without technology.

Rank	Player	Age
1	James Harden	29
2	Paul George	28
3	Giannis Antetokounmpo	24
4	Joel Embiid	24
5	Stephen Curry	30
6	Devin Booker	22
7	Kawhi Leonard	27

Table 8.13 (source: https://www.basketball-reference.com/leagues/NBA_2019_leaders.html)

Rank	Player	Age
8	Kevin Durant	30
9	Damian Lillard	28
10	Bradley Beal	25

Table 8.13 (source: https://www.basketball-reference.com/leagues/NBA_2019_leaders.html)

8. Using the data in “MBB2019” (https://openstax.org/r/Chapter8_Data-Sets), find the mean, median, and mode of the number of free throws made by each team (FT).

Range and Standard Deviation

9. Using the data in “MBB2019” (https://openstax.org/r/Chapter8_Data-Sets), find the range and standard deviation of the number of free throws made by each team (FT).

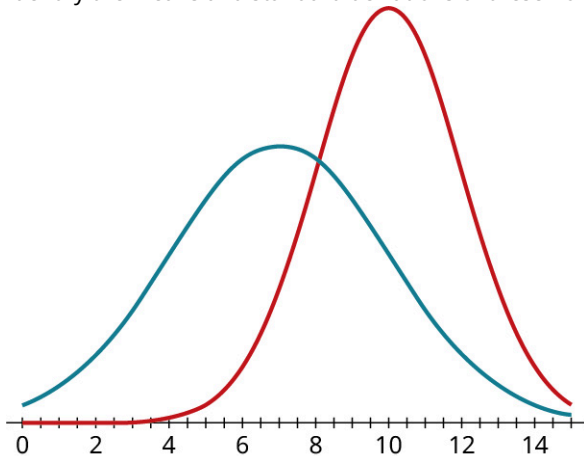
Percentiles

Use the data in “MBB2019” (https://openstax.org/r/Chapter8_Data-Sets), to answer the following:

10. What number of free throws (FT) is at the 40th percentile?
11. At what percentile is Syracuse University, which made 480 free throws?

The Normal Distribution

12. Identify the means and standard deviations of these normal distributions:



Answer the following about data that are distributed normally with mean 200 and standard deviation 20:

13. What proportion of the data are between 160 and 240?
14. What's the standardized score of the data value 235?
15. At what percentile would the data value 187 fall?
16. What data value would be at the 90th percentile?

Applications of the Normal Distribution

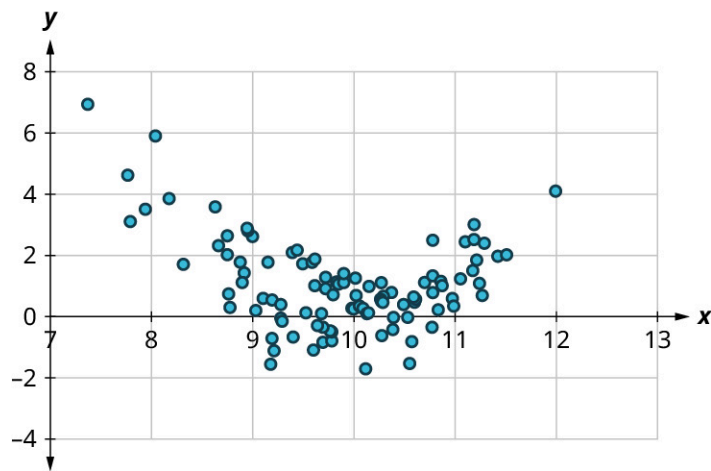
Use the data in “MBB2019” (https://openstax.org/r/Chapter8_Data-Sets) to answer the following:

17. What is the standardized score for Purdue University's number of free throws (FT = 461)?
18. At what percentile would we estimate Purdue's value of FT to fall, using the normal distribution?
19. At what percentile does Purdue's value of FT actually fall?

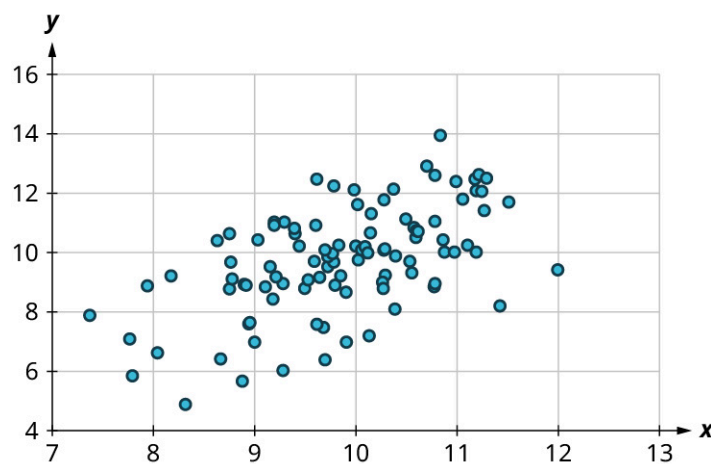
Scatter Plots, Correlation, and Regression Lines

For each of the following scatter plots, decide whether linear regression would be appropriate. If it is, classify the strength and direction of the relationship.

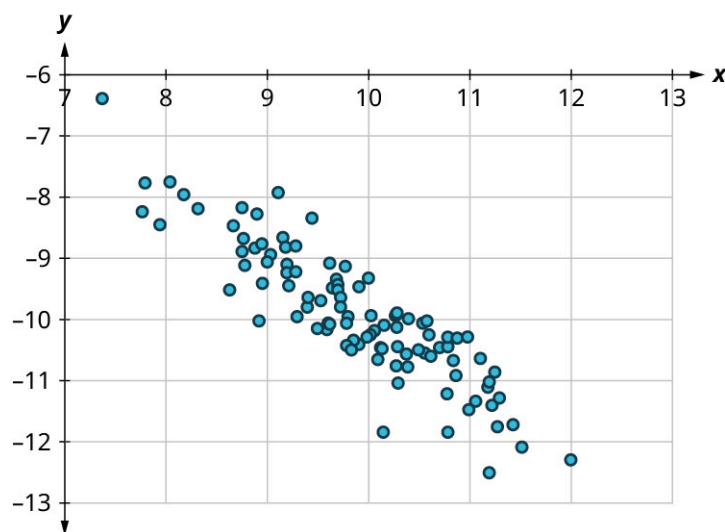
20.



21.



22.



Use the data in “NBA2019” (https://openstax.org/r/Chapter8_Data-Sets) to answer the following:

23. What is the correlation coefficient between the number of field goals attempted per game (FGA) and number of points scored per game (PTS)?
24. What is the equation of the regression line we would use to predict PTS from FGA?
25. Write a sentence to interpret the slope of the regression line.
26. Predict the number of points per game for a player who attempts 16 field goals per game.
27. Danilo Gallinari attempted 13 field goals per game, and averaged 19.8 points per game. Did he score more or fewer points per game than expected? By how much?

Chapter Test

Decide whether randomization is being used in the selection of these samples. If it is, identify the type of random sample (simple, systematic, cluster, or stratified).

1. A vinyl record dealer is trying to price a large collection she's thinking of buying. She looks at every tenth record on the shelf and notes the value.
2. A court clerk is charged with identifying one hundred people for a jury pool for upcoming legal hearings. He has an alphabetized list of registered voters in his jurisdiction, so he uses a random number generator to pick one hundred names from the list.
3. A state agricultural officer is worried about the spread of a parasitic disease among cattle. He chooses 30 cattle farms at random and tests each cow on those 30 farms for the disease.
4. A sample of donations to a blood bank contained these blood types:

A	O	A	O	A	A	AB	O
O	A	AB	A	O	O	A	A
O	O	B	O	B	O	O	O
B	A	O	A	O	O	A	A

Create a frequency distribution for these data.

5. Create a bar chart without technology to visualize the data in question 4.
6. Use the data in "MBB2019" (https://openstax.org/r/Chapter8_Data-Sets) to create a histogram of the number of total rebounds by each team (TRB).
7. The table below shows the number of players on the rosters of each English Premier League team. Find the mode, median, and mean without technology.

Team	Active Players	Team	Active Players
Manchester United F.C.	27	Newcastle United F.C.	28
Manchester City F.C.	24	Aston Villa F.C.	24
Chelsea F.C.	25	Fulham F.C.	27
Arsenal F.C.	28	Southampton F.C.	24
Tottenham Hotspur F.C.	27	Wolverhampton Wanderers F.C.	22
Liverpool F.C.	30	Brighton & Hove Albion	31
Crystal Palace	29	Burnley F.C.	22
Everton F.C.	23	West Bromwich Albion F.C.	26
Leicester City	25	Leeds United F.C.	25
West Ham United F.C.	24	Sheffield United F.C.	27

Table 8.14 (source: https://fbref.com/en/comps/9/Premier-League-Stats#all_stats_squads_standard)

8. Using the data in “MBB2019” (https://openstax.org/r/Chapter8_Data-Sets), find the mean, median, and mode of the total number of rebounds made by each team (TRB).
9. Using the data in “MBB2019” (https://openstax.org/r/Chapter8_Data-Sets), find the range and standard deviation of the total number of rebounds made by each team (TRB).

Use the data in “MBB2019” (https://openstax.org/r/Chapter8_Data-Sets) to answer the following:

10. What number of total rebounds (TRB) is at the 3rd quartile?
11. At what percentile is the University of Evansville, which had 1095 total rebounds?

Answer the following about data that are distributed normally with mean 50 and standard deviation 5:

12. What proportion of the data are between 35 and 65?
13. What's the standardized score of the data value 37?
14. At what percentile would the data value 58 fall?
15. What data value would be at the 40th percentile?

Use the data in “MBB2019” (https://openstax.org/r/Chapter8_Data-Sets) to answer the following:

16. What is the standardized score for the number of total rebounds (TRB) recorded by Mississippi State University (TRB = 1215)?
17. At what percentile would we estimate Mississippi State's TRB value fall, using the normal distribution?
18. At what percentile does Mississippi State's TRB value actually fall?

Use the data in “NBA2019” (https://openstax.org/r/Chapter8_Data-Sets) to answer the following:

19. What is the correlation coefficient between the number of two-point field goals made per game (2P) and number of points scored per game (PTS)?
20. What is the equation of the regression line we would use to predict PTS from 2P?
21. Write a sentence to interpret the slope of the regression line.
22. Predict the number of points per game for a player who makes 6 two-point field goals per game.
23. J.J. Redick made 2.8 field goals per game, and averaged 18.1 points per game. Did he score more or fewer points per game than expected? By how much?